Applied Geography 126 (2021) 102382

Contents lists available at ScienceDirect

Applied Geography

journal homepage: http://www.elsevier.com/locate/apgeog

Developing a geospatial data-driven solution for rapid natural wildfire risk assessment

Bishrant Adhikari, Chen Xu *, Paddington Hodza, Thomas Minckley

University of Wyoming, USA

ARTICLE INFO

Keywords:
Wildfire simulation
Data-driven
Spatial models
Apache spark
Monte Carlo

ABSTRACT

Computational natural wildfire simulation is a computing-intensive process. The process is also challenging because of the need to integrate data with wide spatial and temporal variability. Our study sought to simulate rapidly spreading natural wildfire with fidelity and quality through computational realization. We developed a novel probabilistic wildfire risk assessment tool whose operation was driven by real-time wildfire observations. A Gaussian transformation incorporating present and historical geographical data to the wildfire model was adopted to accommodate scale differences in the datasets. The model outputs, therefore, depict possible spread pathways using Monte Carlo simulations. We created a computational solution for deploying wildfire simulations to a highly scalable, distributed and parallel computing framework, which facilitated a fairly linear increase in the simulation run time as the computation load increased exponentially. Our computational solution synthesized and fully automated the various stages of the process, from data preparation to analysis and visualization. The platform can potentially provide real-time decision-making support to wildfire hazard management.

1. Introduction

Wildfires can undergo large spatial changes over short periods of time, presenting challenges when seeking to identify and report the nature and degree of fire risks to the public. As the risk of wildfires grows in urban-wildland interfaces, communities are becoming more interested in assessing their residences' potential risk levels to wildfire. People can currently turn to authorities charged with monitoring wildfires for categorical information about local wildfire risk levels. However, categorical information has limited actionable value to individuals since it typically applies generalities to large regions designated, for example, as 'high-risk.' For wildfire risk information to be useful and actionable, it should communicate relative and targeted risk assessment based on an individual's actual geographical location (Lion et al., 2002).

Numerous wildfire models have been developed to assist in conducting fire risk assessments at broad scales. However, many models are difficult to implement in real-time because such exercises typically require considerable data preparation and processing effort as well as in-depth knowledge and experience with computational processes. This creates a gap between scientific knowledge and operational solutions for wildfire risk assessment. Wildfire propagation mechanisms have been depicted by using well-defined numerical models; many geographic datasets needed in executing computational wildfire simulations, such as weather forecasts, are readily available in downloadable and useable formats. It is feasible to ease the operational complications by leveraging a workflow process that automates the simulation operation from data preparation to result presentation.

Over the years, fire modeling has evolved from theoretical to empirical. Several wildfire-spread models integrating physical models, empirical observations, and mathematical equations have been deployed to predict the dynamic nature of how wildfires develop and grow on the physical landscape (Sullivan, 2009). Many of these models are deterministic and make fixed value predictions about wildfire spread speeds at a point location based on a combination of weather conditions, fuel status, and a terrain model for a foreseeable period of time. Of these datasets, only terrain may be viewed as a constant in a wildfire behavior model. Weather forecasts are conversely predictions for a foreseeable period of time based on measurements at a given time. As time since observation increases, actual conditions, e.g., instantaneous humidity and fuel condition, vary in a wider interval of values centered on the observation. Meanwhile, an ongoing wildfire could initiate a positive feedback dynamic process where weather conditions affect fuel status introducing further variation in fuel datasets. Combined heterogeneous conditions significantly impact wildfire behavior, but how to

* Corresponding author.
E-mail address: cxu3@uwyo.edu (C. Xu).

https://doi.org/10.1016/j.apgeog.2020.102382
Received 18 November 2019; Received in revised form 5 October 2020; Accepted 17 December 2020
Available online 25 December 2020

incorporate the effect of variations within input data in a deterministic wildfire model remains a challenge. First, we need to define a variation function for assessing the variability possessed by the data sources; and second, we need to develop a process to integrate the variation function with the wildfire model.

Through years of laboratory studies, many wildfire behavioral models have achieved a state-of-the-art form (Thompson et al., 2012). It is feasible to base computational models on these behavioral models in rapid wildfire risk assessment projects. In this study, we used a Gaussian transformation process to generate a series of possible environmental conditions based on the most recent measurements off the ground and estimate data variability by using historical datasets of the same geographic area. The method assumed that geographic realities confine environmental conditions such as temperature and wind, and we could design a variance estimator by analyzing historic environment data. Thus, we provide a solution to the first need, as aforementioned.

Here, we examine how rapidly spreading natural wildfire can be simulated with fidelity and quality through computational realization. We develop a Monte Carlo based computational solution to integrate the Gaussian transformation process into a deterministic wildfire model. Thus, we present a solution to the second need as aforementioned. Details of the integration method are outlined in section 4. Developing our own wildfire model allowed us to explore the necessary computational optimizations otherwise challenging to execute in existing systems like FARSITE and BEHAVE, which are discussed in the next section. Many current wildfire simulation platforms require manual data entry, have long computational timeframes, and do not account for the uncertainties present in weather and fuel moisture data during wildfire spread simulation. To address these issues, we designed an automatic system that can facilitate swift decision-making regarding resource allocation (i.e., firefighting resources) and notifying citizens (i.e., the general public) about potential wildfire risk. We greatly simplify the operational complexity of conducting computational wildfire simulation with a well-designed automated workflow.

We contribute to spatial wildfire modeling by: (1) developing and integrating a Gaussian transformation process which estimates potential data variations in inputs that may affect a wildfire model's outputs; (2) demonstrating a computational workflow, which synthesizes and fully automates the stages of the simulation from data preparation to analysis and visualization; and (3) creating a computational solution for deploying wildfire simulations to a highly scalable, distributed and parallel computing framework which facilitates a fairly linear increase in the simulation run time as the computation load increases exponentially.

2. The ongoing challenge to cope with uncertainty in wildfire spread simulation

Wildfire spread simulation refers to the use of computer-based models to predict how a wildfire might spread (Petrasova et al. 2018). The primary prediction parameters of wildfire simulation models are the rate of spread and its effect, which results in the growth of the fire perimeter. Environmental conditions are weighted into the space-based model for estimating the probability of ignition at different points of the fire front (Bachmann & Allgöwer, 2001). Wildfire risk is commonly determined by the location(s) where the fire is most likely to spread from one or more ignition (Hardy, 2005).

Fuel and weather parameters are key inputs to wildfire-spread simulation. Vegetation cover is crucial in developing fuel models used to estimate fuel dynamics, while environmental and weather parameters are applied to those fuel models to predict fire behavior (Rothermel, 1972). The basis for most fire models comes from Rothermel (1972) and Wagner (1977). Rothermel's surface fire model performs reasonably well on varied landscapes and vegetation covers (Andrews, 2010; Finney, 1998). Under favorable conditions that increase fire intensity and severity, a surface fire may propagate into tree canopies using ladder fuels. The crown fire model developed by Wagner (1977) has been used in most wildfire simulation software and found to closely approximate crown fire behavior in the United States (Finney, 1998). As with other wildfire models, these models are physical or empirical models, which materialize observed wildfire behaviors as numerical equations (Rothermel, 1972).

Efforts to represent and model wildfire spread with high fidelity and quality have received considerable attention and evolved over the years. First, Wagner (1977) added the capability to simulate crown fire to Rothermel's surface fire model. Finney (1998) combined Wagner and Rothermel's models modifying the integration of surface and crown fire propagation developed by Wagner (1977) by incorporating crown base height, crown bulk density, fire-line intensity, and fuel characteristics to estimate fire intensity along with the transition of surface fire to crown fire. The integrated model used a threshold based on the rate of spread, canopy characteristics, and fire-line intensity to switch between surface fires and crown fires.

With natural fires, fuel conditions change gradually as wildfire progresses. Albini (1976) proposed that combining 1, 10, and 100-h dead fuel moisture into a single value can allow a time-based burnability prediction to be derived as a wildfire progresses. Scott and Burgan (2005) devised an approach for deriving fuel bed depth, heat content, and surface-area to volume ratios from fuel model parameters. Rothermel's commonly used wildfire model expects these parameters. The refinements have all been coded into wildfire simulations widely used by researchers and practitioners in the field. For example, FARSITE is a spatially explicit fire growth and simulation platform widely used by federal agencies dealing with wildfire suppression and management in the US. FlamMap was developed by USFS to incorporate the spatial variabilities in fuel moisture and wind conditions, which were not wholly included in FARSITE (Finney, 2006). FSPro is a GIS-based probabilistic simulator that estimates the long-term fire growth (more than five days) (Noonan-Wright & Erin, 2011). In addition to these mostly used computational wildfire platforms, others have been created to support various purposes such as BehavePlus and WIFIRE (Altintas et al., 2013; Andrews et al., 2005). BehavePlus was not designed to be a complete system for simulating wildfire across a landscape, whereas WIFIRE is still in the prototype phase and requires a considerable workforce, sensor networks, and computation resources to sustain and operate as expected.

Weather conditions, fuel, and terrain interact to define wildfire behavior. Among the three, fuel composition and terrain (slope, aspect) exert a stable force, and weather conditions are a volatile force. Together they form a coupled chaotic system (Chapin et al., 2009) and can frustrate ambitions to attain model precision and accuracy, which define a model's fidelity and quality. As wildfire's behavior is sensitive to environmental variations, the fidelity and quality of different systems in operation is determined by the extent to which they can be calibrated for local geography. FARSITE, for example, was initially developed to study prescribed fires that might be active for weeks or so in national parks or wildlands (Finney, 1994). Because it is spatially and temporally explicit and uses now readily accessible GIS technology to map predictions, FARSITE is a widely implemented platform. However, when the platform was applied in Europe, the calibration process was challenging mainly because fuel characteristics are very different from those in North America (Dimitrakopoulos, 2002; Pastor et al., 2003). Arca et al. (2007) demonstrated that using a custom fuel model and local meteorological data have significant effects on FARSITE's performance when simulating wildfires in Italy.

The majority of existing wildfire behavior models were tested and calibrated in a controlled laboratory environment that often use generalized and simplified artificial conditions, which results in their limited efficiency for landscape level wildfire risk assessments (Gollner et al., 2015). For example, there is considerable uncertainty between predicted and actual burned areas by wildfires because of generalized fuel conditions, especially when positive feedback processes begin to sustain changes in vegetation conditions like structure, moisture content, and fuel type. Thus, wildfire model fidelity and quality outside of the laboratory still requires additional research to enhance the model performance based on natural conditions.

3. New directions in wildfire simulation

Rochoux et al. (2015) argue that a good understanding of various environmental variables is crucial in forecasting wildfires in a reliable manner. Data-driven approaches have demonstrated the capacity to improve wildfire models' quality incrementally by adding computation-based calibration mechanisms to simulation processes. These approaches employ forward modeling to produce forecasts, and the feedback calibration process uses observed wildfire conditions to refine estimated model parameters (Rochoux et al., 2014). They are in line with the Dynamic Data Driven Applications Systems (DDDAS) paradigm, which relies on accumulating large observational datasets to implement Bayesian refinements on multiple model coefficients in wildfire simulation (Madey et al., 2007). Such refinements ensure that a wildfire model's parameters match those of a specific wildfire. The model parameters (i.e. coefficients and constants) can be modified to reflect local effects, such as fuel heterogeneity, topographic anomalies, and artificial features like fire breaks and roads. Data with high spatial and temporal resolutions are increasingly being captured through unmanned aerial vehicles or drones and citizen science to support more reliable wildfire forecasting.

There are many challenges when existing wildfire models are used to assess the risks of active wildfires. First, the deterministic nature of wildfire models prescribes a one-to-one relationship between data inputs and model outputs. Thus, regardless of the model's fidelity and quality, simulations are restricted by data availability due to data collection approaches dependent on traditional remote sensing technologies (Leblon et al., 2012). Therefore, the reliability of the outputs from even high-quality wildfire models diminishes as the time since the last measurements increases. However, while conditions between field measurements are uncertain, environmental status variability, e.g., weather conditions, can be derived as historical observations are available. For example, the FSPro uses historical weather data to create artificial weather conditions based on stations proximal to a wildfire and leverages the stations' terrain factors to extrapolate seasonal variations. The model then forecasts wildfire conditions based on artificial weather conditions (US Geological Survey 2015). FSPro targets the needs of long-term decision-making (more than five days) (Noonan-Wright & Erin, 2011). Effective prediction of the progression of a wildfire over multiple days using FSPro is affected by the built-in assumption of constant weather and fuel conditions, potentially leading to significant discrepancies between the statuses of short-term forecasted and observed wildfire.

Our study uses historical data for an active wildfire study area to define the distribution of possible model input values between field observations. While simulating the wildfire's spread behavior, we assumed a normal distribution (i.e., Gaussian distribution) of weather and fuel moisture with the mean around current observed data and variance calculated based on the historical data. To integrate the Gaussian transformation function into the deterministic wildfire model, we used a Monte-Carlo approach. This helped us to generate a series of wildfire propagation paths that had a one-to-one relationship with a series of model inputs randomly generated by the Gaussian function. Gaussian distributions for weather and fuel moisture are generated and updated for each hour of simulation to account for any fluctuations during the duration of the simulation. This helped address one of the significant limitations of FSPro concerning the constant weather and fuel conditions.

Table 1
Temporal and spatial data inputs used in the wildfire computational simulation.

Dataset			Resolution	
Name	Source	Size	Spatial	Temporal
HRRR	NOAA	11.5 GB	3 Km	1 Hour
VIIRS active fire hotspot	NASA	≈ 20 MB	375 m	12 Hours
MODIS active fire data	NASA	≈ 20 MB	1 km	1 Day
MODIS surface reflectance	NASA	≈ 3 GB	500 m	1–2 Days
Dead fuel moisture	USFS	≈ 1 GB	2.5 Km	1 Day
Digital elevation model	USGS	26 GB	10 m	2017
Vegetation and fuels	Landfire	3.5 GB	30 m	Jan 2014
Hist. wildfire perimeter	GeoMAC	100 MB	N/A	24 Hours
Historical HRRR data	U. of Utah	9 GB	3 Km	24 Hours

4. Methods

4.1. Monte Carlo based wildfire risk assessment and GIS realization

Our model represents a phenomenological interpretation of wildfires that considers how they spread across the landscape as a series of ignitions. Fons (1946) articulated the theory behind the surficial spread of wildfires while Rothermel formulated the mathematical to describe this process (Andrews, 2018).

$$R = (I_R \xi\ (1 + \phi_w + \phi_s)) / (\rho_b \varepsilon Q_{ig})$$

Where R is the rate of spread in meters/minute, I_R is the rate of energy release per unit area of the fire front, ξ is the propagating flux ratio for igniting adjacent fuel particles, ϕ_w is the wind factor, ϕ_s is the slope factor, and ρ_b is fuel bed factor, ε is the effective heating number of the fuel bed, and Q_{ig} is the amount of heat required to ignite one pound of fuel. $I_R\xi$ describes the amount of heat released at no-wind and no-slope conditions. $(1 + \phi_w + \phi_s)$ calculates the increase due to the wind and slope conditions. Together $I_R\xi\ (1 + \phi_w + \phi_s)$ produces the propagating flux, and $\rho_b \varepsilon Q_{ig}$ defines the heat required to ignite the fuel. The model depicts scenarios where there is a process for wildfire spread by which energy released from a fire source gradually heats and ignites its surroundings. The areal extent of ignitions depends on Rothermel's model.

Within the wildfire risk assessment domain, a probabilistic approach over a deterministic approach is preferred to address uncertainties in wildfire spread simulation due to fluctuations in weather, topography, and fuel conditions (Bachmann & Allgöwer, 1999; Thompson et al., 2012). A Monte-Carlo based simulation has been used to minimize biases in deterministic wildfire spread models and estimate relative wildfire risk based on the number of times the simulated wildfire perimeters have burned an area (Thompson et al., 2012). For example, in a 100 Monte Carlo runs simulation, a burnt area predicated by 60 runs has a higher risk level than a burnt area predicted by 10 runs. Regarding actual data transformation, fuel moisture is calculated from integrating average boundary conditions based on previous year's observed values, hours of rain, and daily temperature range. The fuel moisture value used in the model is expressed as a percentage; hence can approximate a normal distribution. Weather conditions are generated by a Gaussian function with a field measurement as the mean value and the variance calculated from historical data. These normal distributions were considered independent of each other for the purpose of this research to evaluate all possible combinations. The spatial autocorrelation among weather variables (wind speed, direction, and temperature) and fuel moisture is considered by using the reaction intensity in the wildfire spread model developed by Rothermel (1972).

4.2. Data

Data inputs to the wildfire simulation process are presented in Table 1 together with their sources, spatial and temporal resolutions, and approximate local storage space.

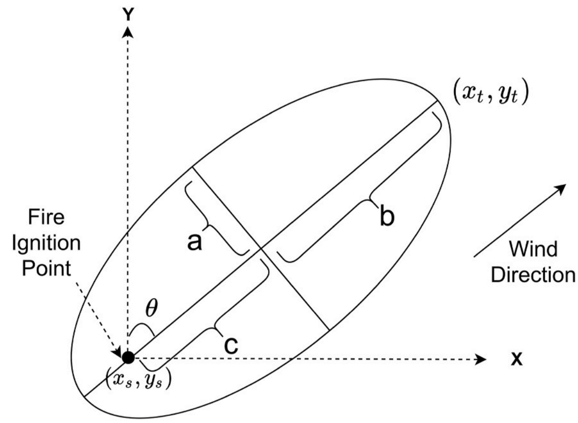

Fig. 1. Elliptical fire spread.

Data for this study have different spatial and temporal scales and different update timings. Datasets such as HRRR, dead fuel moisture, vegetation and fuels, historical wildfire perimeter, and historical HRRR data are available only for the United States, whereas datasets such as VIIRS active fire data, MODIS active fire data, MODIS surface reflectance, and Digital Elevation Models (DEMs) are global. All datasets except the historical wildfire perimeter derived from GeoMAC are in raster format. Current and forecasted weather data are obtained from the High-Resolution Rapid Refresh (HRRR) model data published by the National Oceanic and Atmospheric Administration (NOAA). Although HRRR, vegetation, and fuels datasets are only available for the United States, they could be substituted for weather data extrapolated from ground-based observation and vegetation along with fuel moisture data generated by classifying MODIS and Landsat data for areas outside of the United States. The data are updated every hour and weather forecasts are provided for up to 18 h. Relative humidity, air temperature, and wind values are used to calculate the wildfire spread rate and potential direction. HRRR data older than two days are not available from the NOAA website and so to generate the stochastic weather function, *Pando Archive* an HRRR archive maintained by the University of Utah was used. Active wildfire hotspot data were obtained from the GeoMAC Wildfire Application, which manages active fire data produced by NASA's VIIRS and MODIS platforms. Historical wildfire perimeter data were acquired from GeoMAC. Vegetation data are available from the LANDFIRE Program, and the data were used for estimate surface and crown fire intensities. The daily dead fuel moisture data by Wildland Fire Assessment System (WFAS) from the US Forest Service was used to calculate fire intensity and flammability of fuels. The dead fuel moisture data (1, 10, and 100-h) are extracted from the WFAS's National Fire Danger Rating system. The estimation of live fuel moisture is performed by using the surface reflectance data captured by the MODIS Aqua and Terra satellites. Topographic data such as elevation are collected from the US Geological Survey.

4.3. The spatialization of wildfire spread

The rate of wildfire spread that is predicted using Rothermel's model can be visualized using elliptical boundaries. Richards (1990, 1995) and Finney (1998) used a simple ellipse with fire ignition point near the ellipse's rear focus to represent the geometry of wildfire spread. The elliptical growth model for angular (θ) wind direction given by Richard's (1990) is schematically represented in Fig. 1.

The ellipse components, a, b, and c are determined from local topography, wind, and fuel conditions (Richards, 1990). The direction of the midflame wind vector is denoted by θ. The elliptical dimensions estimate the fire extent and spread direction for a single point source on a wildfire front. The front line is dissected to unit segments and a vertex on each segment serves as the fire source, following the Huygens wavelet principle, to model the entire wildfire front (Pastor et al., 2003).

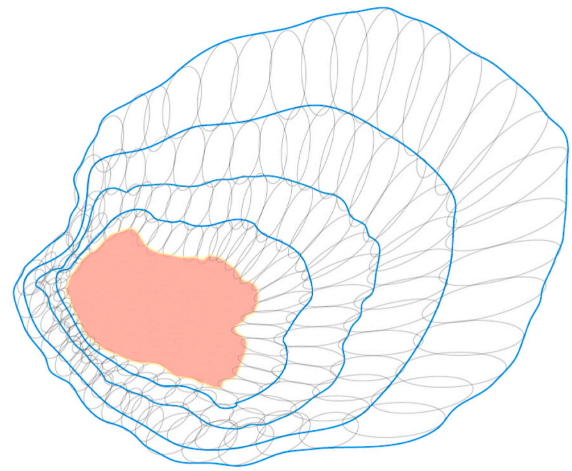

Fig. 2. Wildfire spread based on Huygens principle and elliptical firelets (Adapted from Rios, Jahn, and Rein 2014).

The Huygens wavelet principle (Fig. 2) is used in several vector-based simulation systems, such as FARSITE, to propagate fire; primarily based on wind direction. The resolution of the model run determines the number of smaller segments. At each vertex, the spread extent of an ellipse is determined by the spread rate calculated at the vertex. The red area indicates the fire start area and is divided into smaller segments with each segment as a point source of ignition. The spread was simulated for each source using ellipses represented in gray to calculate a new fire perimeter at the next step, represented by blue polygons. The combination of small ellipses into a fire perimeter can result in self-intersection and crossovers. These geometrical distortions are solved by establishing topological rules and calculating the polygons' convex hull at runtime.

4.4. The production of the wildfire risk map

Producing a wildfire risk map begins by specifying the following elements: 1) total prediction hours, 2) current wildfire boundary, 3) spatial resolution of the simulation, which determines the number of segments of a fire boundary, and 4) the total number of Monte Carlo runs in each simulation step. The values of weather factors (i.e., wind, temperature, and humidity), topography (elevation, aspect, slope), and fuel (fuel class, fuel moisture) are extracted from their respective raster files to calculate the rate of spread for ellipses at individual segments. Where the rate of spread is zero, the fire propagation is stopped for that segment. A topographical correction is applied when the rate of spread is greater than zero. The final perimeter for each time step is calculated by generating the convex hull of the ellipses. The simulation process runs until the specified total prediction hour and the number of Monte Carlo runs are reached. The weather and fuel moisture values are generated stochastically for each Monte Carlo run such that the predicted boundaries reflect variations because of weather and fuel moisture uncertainties.

A set of predicted boundaries, determined by the number of Monte Carlo runs, are produced in each simulation step. All Monte Carlo predictions are synthesized based on Thompson et al.'s (2012) approach, which estimates wildfire risk using the number of times the simulated wildfire perimeters over an area. A GIS-based overlay operation is used to integrate all the layers representing the predicted scenarios. Fig. 3 shows the Keystone fire simulation (July 3, 2017, Albany County Wyoming), which maps the predicted fire-risk perimeter 30 h after the initial ignition. Monte-Carlo simulation for this research was performed for 20 runs due to a limitation in available hardware and computation resources. Here, the dashed red line represents the observed fire

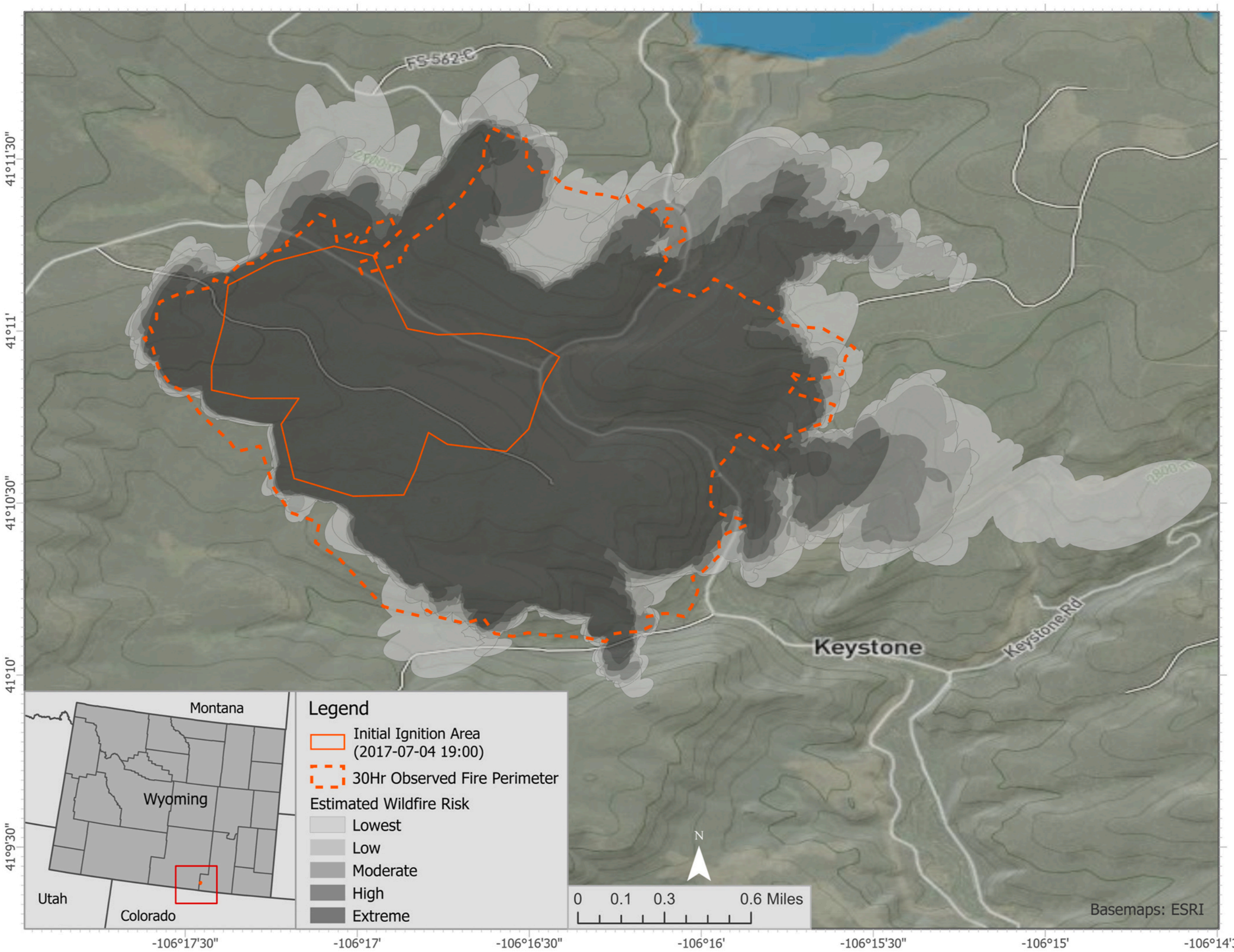

Fig. 3. Mapping wildfire spread risk of the Keystone fire, July 3, 2017.

boundary 30 h after its ignition (shown by solid red line), and the predicted risk levels (shown by solid gray fills) result from the spatial overlay of all Monte Carlo runs. It can be observed that the high-risk area (the dark gray area) covers most of the actual burned area. More importantly, at the edge of the wildfire, where uncertainty is expected to be the greatest, shows that several individual model runs predicted the likelihood of burning. The observed boundary validates the predicted extent of the wildfire risk as shown by the semi-transparent gray areas in Fig. 3. Wyoming was chosen as a case study area because of the state's growing wildfire problem and readily available wildfire data as well as the researcher authors' experience in this area.

4.5. Computational optimization

To achieve rapid simulations that could be useful in risk assessments, we deploy the implementation platform using Apache Spark, a distributed and parallel computing framework. The GIS datasets are stored in a distributed manner across multiple machines using the Hadoop Distributed File System (HDFS). HDFS stores the copy of data in more than one computer to allow simultaneous access from multiple computers with minimal network latency (Borthakur. 2008).

The PostgreSQL relational database with PostGIS extension is used to store and process intermediate simulation results for spatial data retrieval efficiency (e.g., simulated perimeters). PostgreSQL database is only installed on the master node with read/write permissions assigned

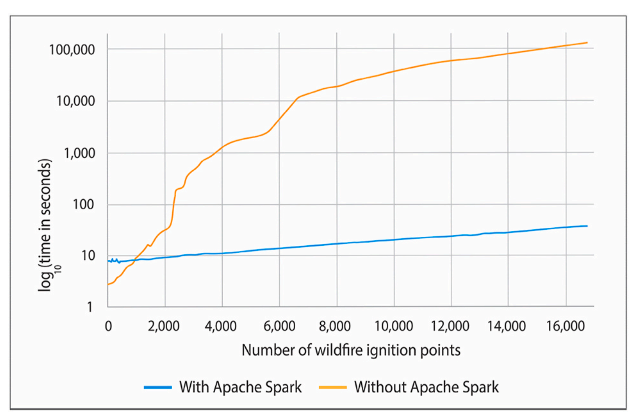

Fig. 4. Computing time with and without Apache Spark.

to the five worker nodes. This centralized PostgreSQL database is scalable and supports easy addition or removal of worker nodes as appropriate without modifying the database's configuration.

The default configuration of Apache Spark was not considered efficient for computational resource management (Sun et al., 2016). Configurations are optimized at runtime and applied to constrain the computational resources used by individual Apache Spark instances on

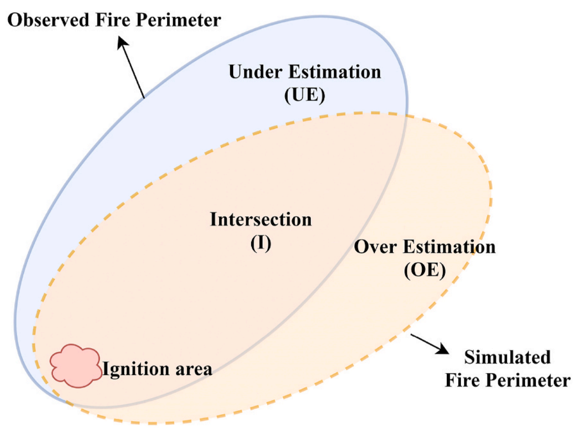

Fig. 5. Representation of observed and simulated fire perimeter Adapted from Duff et al. (2016).

worker nodes. Apache Spark configurations such as maximum allocated memory, memory overhead, the number of computer cores, and worker machines are adjusted as needed for each simulation. Inputs from users concerning wildfire simulation tasks are used to estimate the amount of memory required to run the simulation efficiently. The dynamic allocation functionality of Apache Spark is used to automatically add or remove spark executors based on the current workload and available resources. Additional scripts using PySpark were developed to ensure the resources are released as soon as the process is completed or if the spark job had been idle for a long time.

Fig. 4 shows that for <1000 simulated ignition points, the computation time required using Apache Spark is more than the traditional in-memory computing approach without Apache Spark. The quality of our approach relies on a sufficient amount of runs per simulation. A small overhead in computation time while using Apache Spark can significantly increase the overall execution time when multiplied by the number of model runs. The simulated ignition points here represent the individual points on the outer surface of the fire perimeter, which would act as a point source that propagates the wildfire in the model.

For this study, the traditional standalone computer-based computation is used whenever the number of ignition points is < 1,000, automatically switching to Apache Spark once the number of ignition points crosses the 1000-point threshold. This strategy achieves better computational efficiency than using either one of these approaches alone.

5. Results

We evaluate the performance of the Monte Carlo based wildfire risk assessment. Several statistical indices are determined to compute the

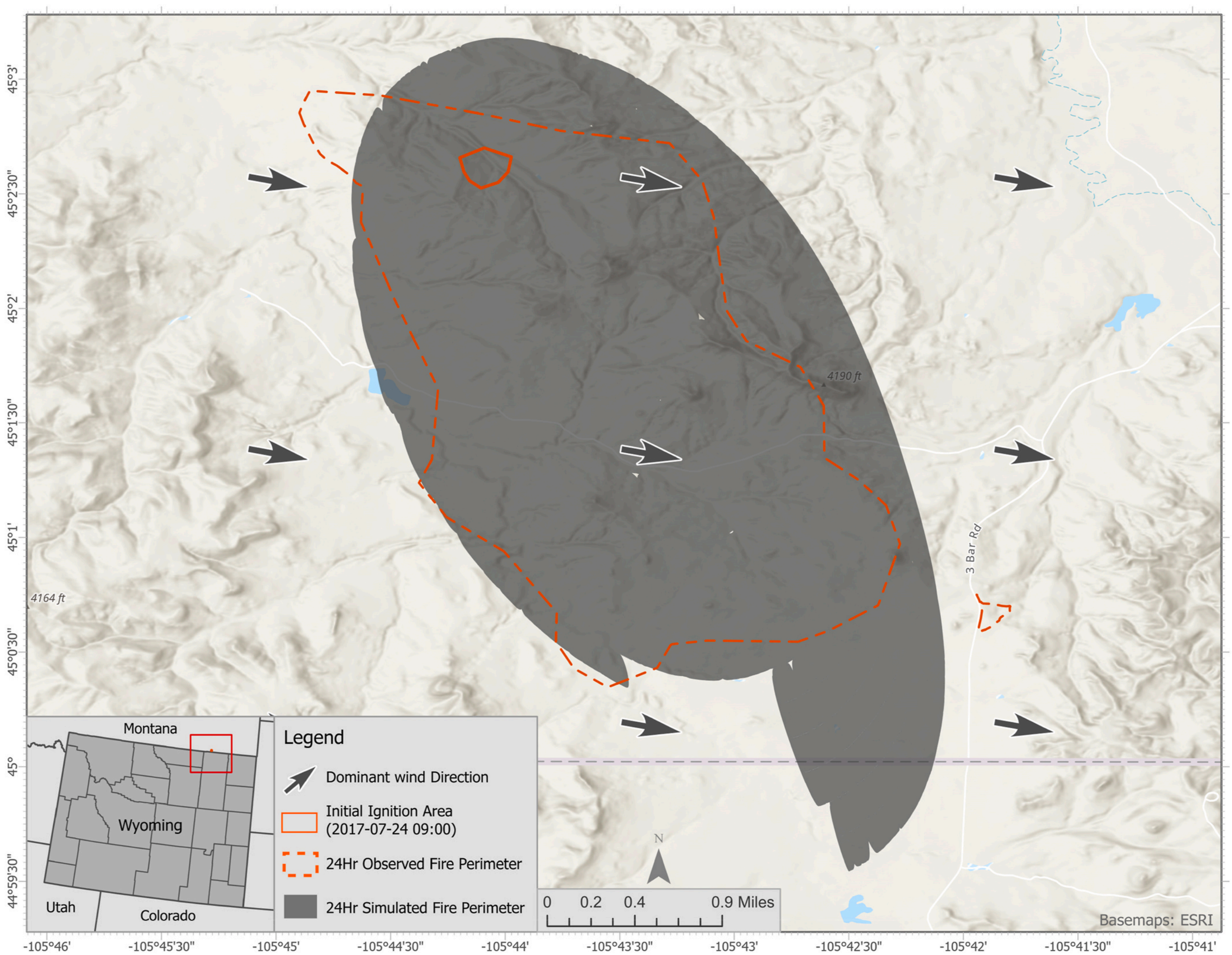

Fig. 6. Predicted (gray fill) and observed (dashed red line) boundaries of the Buffalo fire. The ignition area (solid red line) is minimal in comparison to both observed and predicted fire extents. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

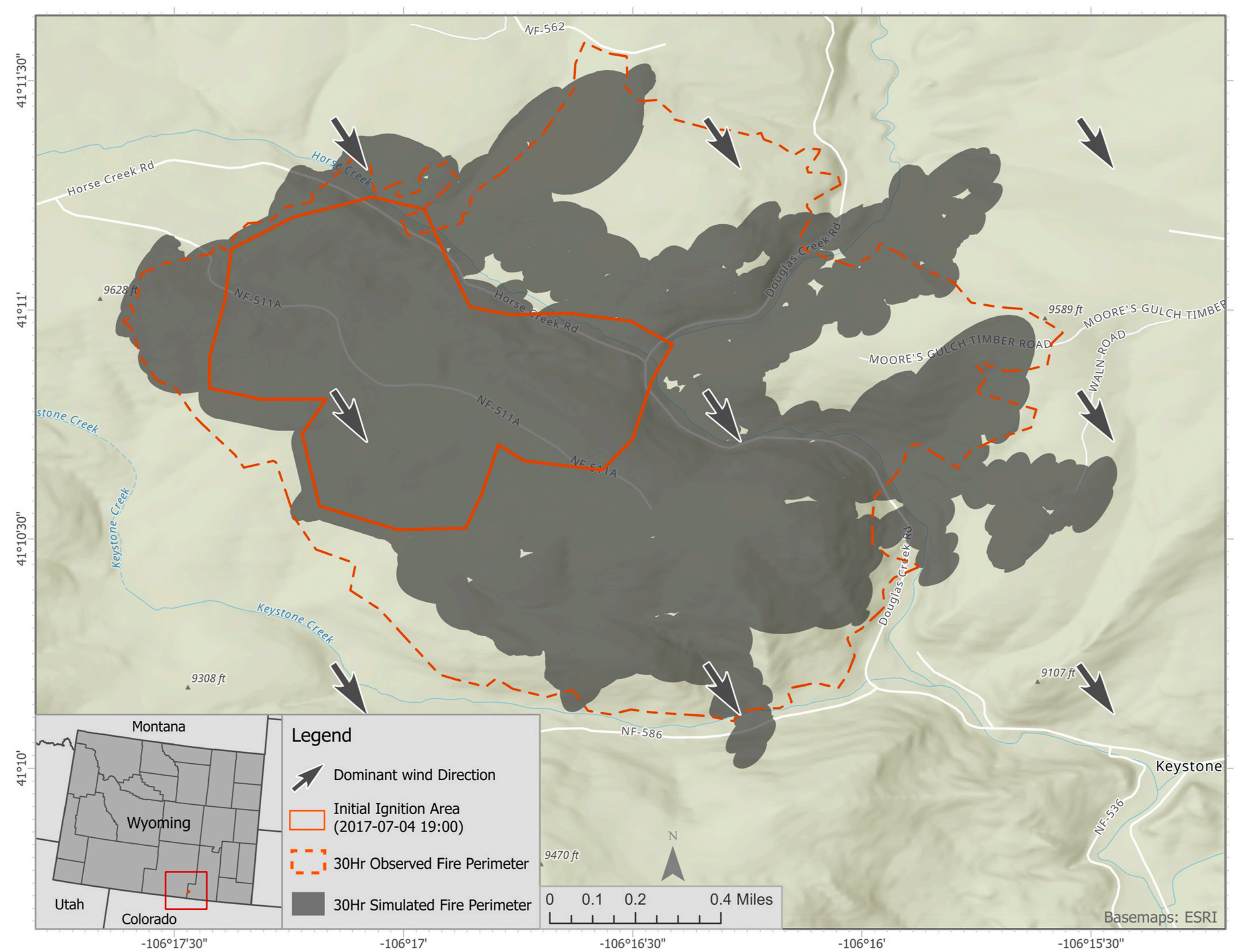

Fig. 7. Predicted (gray fill) and observed (dashed red line) boundaries of the Keystone fire. Here, the ignition area (solid red line) is about half the observed fire extent's size and a third of the predicted fire extent. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

differences between simulated and observed wildfire perimeters. Area Difference Index (ADI) combined with over and under predictions, and F1 score combined with Precision and Recall were the most suitable indices to evaluate a wildfire spread simulation model (Duff et al., 2016; Filippi et al., 2014). Calculating ADI and F1 scores requires observed and simulated perimeters at the same point in time (Duff et al., 2016).

5.1. Area Difference Index (ADI)

The ADI was developed to provide a "simple metric of fire simulation predictive performance," using an index of incorrect estimation as a fraction of the correctly predicted area at a specific time (t) (Duff et al., 2016). The ADI is calculated separately for the under-prediction (ADI_{under}) and over-prediction (ADI_{over}). Fig. 5 presents an illustration of the parameters used in the evaluation.

$$ADI\ (t) = (OE\ (t) + UE\ (t)) / (I\ (t))$$

$$ADI_{under} = UE(t)/I(t)$$

$$ADI_{over} = OE(t)/I(t)$$

5.2. F1 score

The F1 score allows us to measure the accuracy of spread prediction by combining the Precision and Recall of the prediction results. The F1 score is calculated as follows:

$$F1 = 2\ I\ (t) / (I(t) + UE(t) + I(t) + OE(t))$$

Additionally, Precision and Recall are calculated to measure the "specificity of prediction" and "comprehensiveness of a prediction" (Duff et al., 2016). Precision shows the fraction of the correctly predicted area with the total simulated area, whereas recall shows the fraction of the correctly predicted area over the total observed area. Precision and Recall are calculated using the following equations.

$$Precision = I(t) / (I(t) + OE(t))$$

$$Recall = I(t) / (I(t) + UE(t))$$

Three historical wildfires are used to present the results of the evaluation. Figs. 6–8 present Buffalo, Keystone, and Pole Creek fires and the maps show predicted wildfire boundaries (i.e. semi-transparent gray area) and observed (solid red line). The ignition area (i.e., solid red line) was generated from VIIRS data and used as the starting area for the

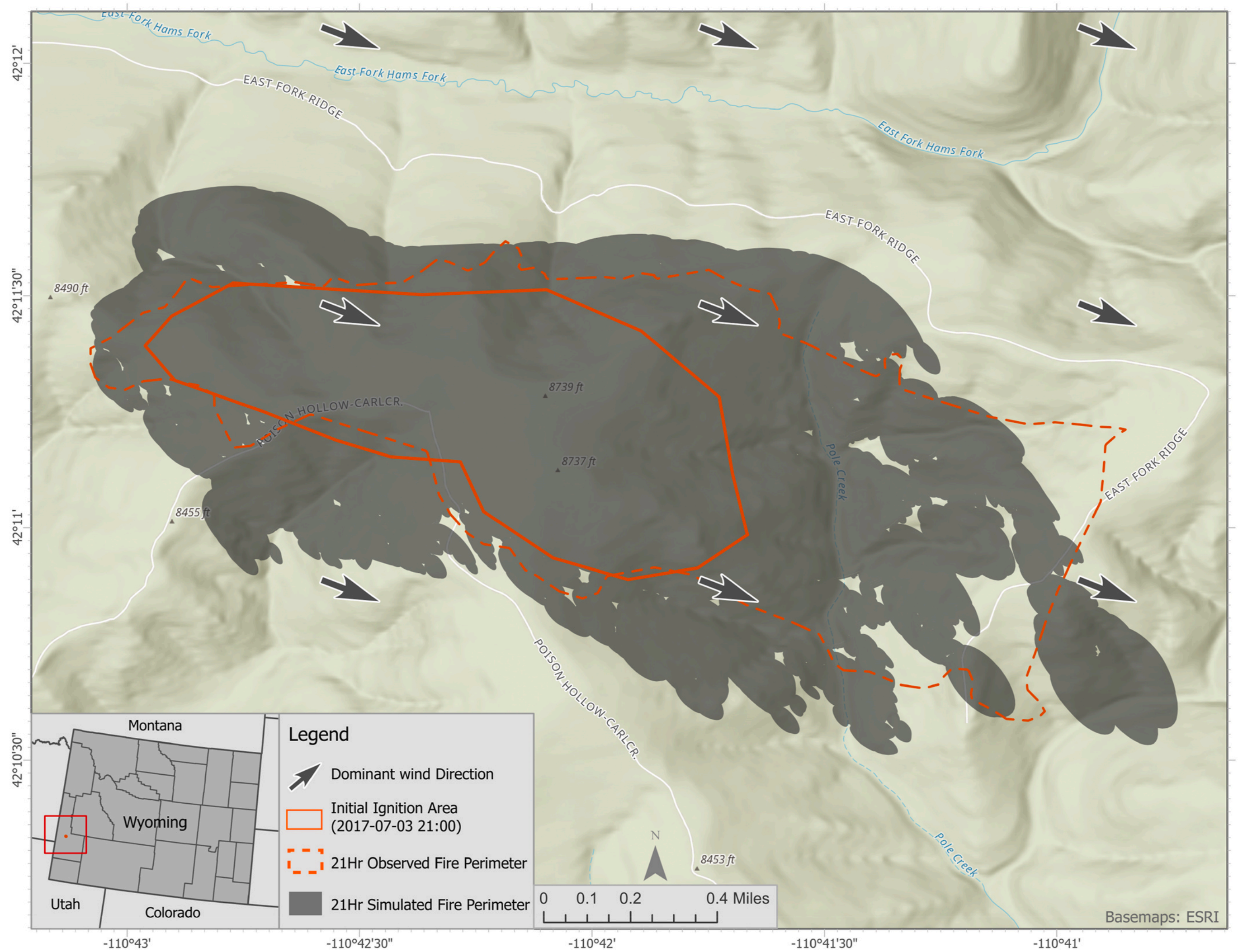

Fig. 8. Predicted (gray fill) and observed (dashed red line) boundaries of the Pole Creek fire. Here, the ignition area (solid red line) is slightly more than half the observed fire extent's size and a roughly half of the predicted fire extent. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Summary of simulated wildfires.

Fire Name	Simulation start date	Simulation end date	Simulation duration (hr)	Observed perimeter source
Buffalo Fire	2017-07-24 09:00	2017-07-25 09:00	24	VIIRS & GeoMAC
Keystone Fire	2017-07-04 19:00	2017-07-06 01:00	30	VIIRS
Pole Creek Fire	2017-08-04 21:00	2017-08-05 18:00	21	VIIRS & GeoMAC

wildfire spread simulation. The gray arrows with white border show the dominant wind direction during the wildfire event.

Here, the observed perimeters from VIIRS and GeoMAC are used to evaluate wildfire spread and risk predictions. The simulation information is summarized in Table 2 and Table 3 shows the statistical measurements.

The simulations revealed that the model overpredicted all the fire perimeters with the Keystone fire having the least amount of over-prediction. Over-prediction was more pronounced in areas with less vegetation cover such as for the Buffalo Fire. It was observed that fire could cross small creeks and water bodies, which might be due to input datasets' coarse resolution. Cases of underprediction in all simulations were relatively minimal compared with overpredictions as shown by the lower values of ADI_{under}.

6. Discussion

This study proposed and then tested a stochastic transformation of

Table 3
Statistical analysis of wildfire spread simulation fidelity and accuracy.

Fire Name	I	UE	OE	ADI	ADI_{over}	ADI_{under}	F1	Precision	Recall
Buffalo Fire	9.09	0.32	4.54	0.53	0.50	0.04	0.96	0.67	0.97
Keystone Fire	3.62	0.84	0.60	0.40	0.17	0.23	0.85	0.86	0.81
Pole Creek Fire	2.54	0.23	1.51	0.69	0.59	0.09	0.85	0.63	0.92

Table 4
Computation time with and without using Apache Spark.

Fire Name	Iteration	Computation time (minutes)		% Improvement
		Without Apache Spark	With Apache Spark	
Buffalo Fire	24	204.6	22.34	915.8
Keystone Fire	30	295.8	28.57	1035.3
Pole Creek Fire	21	123.6	19.66	628.7

weather and fuel conditions derived from current and historical observations. The stochastic transformation was applied in wildfire simulation models to explore the potential of conducting effective near-real time risk assessments during fire events. In this approach, Gaussian-based functions were used to explicitly approximate weather and fuel conditions, which fluctuated based on ground conditions. Our computational engineering approach reduces the total time of simulation cycles from an average of hours to minutes by incorporating parallel and distributed computation technologies.

The primary focus in the wildfire spread simulation is to reduce the degree of underprediction of fire perimeter rather than maximizing accuracy, although overpredictions might result in inefficient use of valuable resources (Zhong et al., 2016). Overprediction of a fire model results in computational metrics that might be viewed negatively. However, predicting fire behavior can have serious consequences in saving property and life. In our qualitative assessment of the model simulations, the degree of overprediction was preferable in terms of risk assessment, if one were to use the model to assess whether people should be evacuated because of an encroaching wildfire. The precision for Keystone Fire was the highest as it had a much lower degree of overestimation, but overall, all model runs suggest a fair degree of accuracy with enough error that could inform the public of imminent risk. Considering the Recall and F1 score, all the simulation tests showed the model performed reasonably well among various terrain and burn conditions.

Using Apache Spark helped us achieve a near-real-time estimation of potential fire perimeter for more extended fire events. Based on how we determined the number of ignition points, or individual fires on the flaming front, once more than 1000 runs in a single simulation step are needed, parallel computation using Apache Spark is more efficient than traditional computation on a standalone computer. With optimizations implemented, the computation time is significantly reduced; and it's achievable to have a rapid risk assessment platform for supporting wildfire management decision-making (Table 4).

Tobler's First Law prescribes that geographic phenomena are not randomly distributed but bounded by spatial locational factors. The existence of spatial dependency implies that local geographical realities influence the weather and fuel conditions. This relationship was captured in statistical models where weather and fuel moisture conditions were randomly generated and incorporated to initiate simultaneous wildfire model runs. A probability surface of wildfire risk distributions was then generated by synthesizing the results of all model runs. Our approach introduced temporally correlated weather variability into the wildfire simulation. This approach relies on current and historical weather observations, but instead of using extrapolation and spatial autocorrelation-based process, it created an interpolated normal distribution of weather forecasts. We use the hypothesis that long-term observations of geographic conditions, e.g., weather conditions, should help determine the possible environmental conditions in a given study area. Although a Gaussian-based distribution was assumed in the study, future work will examine other distribution functions to gauge the effect of implementing different statistical distribution functions. Also, the empirical nature of the approach we used demands multiple experiments to effectively optimize the balance between computational requirements and the wildfire simulation performance, as the real-time simulation is considered an advantage.

7. Conclusions

This study developed a data-driven wildfire simulation model that provided the opportunity to design a new probabilistic-based system for assessing wildfire risk. The Gaussian transformation of weather and fuel conditions suggested a new approach for capturing and accounting for the uncertainties typical in coarse resolution geographic data. The transformation facilitated a Monte Carlo-based wildfire simulation process that explicitly reflected the uncertainties of real-world fire boundaries based on uncertain inputs. The values of ADI and F1 scores suggest that the wildfire simulation's fidelity and quality were improved to considerable extents. Our platform's computational execution is inevitably demanding due to large data size, complex spatial analysis, and the large number of concurrent model runs. The use of Apache Spark, considerably reduced the total computing time from several hours to just a few minutes. Thus, the current model can potentially provide real-time decision-making support to wildfire management. The model ignores the complex coupling relationships among weather, fuel conditions, and terrain, including spatial and temporal autocorrelations between the variables (e.g., between temp and moisture). Future work can implement numerical weather prediction models to couple a wildfire and the atmosphere. Also, an Ensemble Kalman Filter (EnKF) approach will be integrated into the simulation process for calibrating the model's parameters based on new observations. It is currently calibrated for and only useable in the United States. However, it is extensible and can be deployed to other regions by replacing the wildfire model's computational component with one well-calibrated for a target region.

Author statement

We have no conflicts of interest to disclose. Please address all correspondence concerning this manuscript to me at cxu3@uwyo.edu.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apgeog.2020.102382.

References

Albini, F. A. (1976). *Estimating wildfire behavior and effects* (Vol. 30, p. 30). Department of Agriculture, Forest Service, Intermountain Forest and Range Experiment Station.
Altintas, I., et al. (2013). *Wifire: A scalable data-driven monitoring, dynamic prediction and resilience cyberinfrastructure for wildfires* (pp. NH51C–1633). AGU Fall Meeting Abstracts.
Andrews, P. (2010). Do you BEHAVE? - application of the BehavePlus fire modeling system. In D. D. Wade, & M. L. Robinson (Eds.), *Proceedings of 3rd fire behavior and fuels conference; 25-29 October 2010; Spokane, WA* (Vol. 17). Birmingham, AL: International Association of Wildland Fire.
Andrews, P. L. (2018). The Rothermel surface fire spread model and associated developments: A comprehensive explanation. In *Gen. Tech. Rep. RMRS-GTR-371* (Vol. 121, p. 371). Fort Collins, CO: US Department of Agriculture, Forest Service, Rocky Mountain Research Station.
Andrews, P. L., Bevins, C. D., & Seli, R. C. (2005). BehavePlus fire modeling system, version 4.0: User's Guide. In *Gen. Tech. Rep. RMRS-GTR-106 revised* (Vol. 132, p. 106). Ogden, UT: Department of Agriculture, Forest Service, Rocky Mountain Research Station.
Arca, B., et al. (2007). Evaluation of FARSITE simulator in a Mediterranean area. In *Proceedings of the 4 th international wildland fire conference*. Seville, Spain.
Bachmann, A., & Allgöwer, B. (2001). A consistent wildland fire risk terminology is needed. *Fire Management Today, 61*(4), 28–33.
Borthakur, D. (2008). HDFS architecture guide. *Hadoop Apache Project, 53*(1–13), 1–13.
Chapin, F. S., Folke, C., & Kofinas, G. P. (2009). A framework for understanding change. In *Principles of ecosystem stewardship* (pp. 3–28). New York, NY: Springer.
Dimitrakopoulos, A. P. (2002). Mediterranean fuel models and potential fire behaviour in Greece. *International Journal of Wildland Fire, 11*(2), 127–130.

Duff, T. J., Chong, D. M., & Tolhurst, K. G. (2016). Indices for the evaluation of wildfire spread simulations using contemporaneous predictions and observations of burnt area. *Environmental Modelling & Software, 83*, 276–285.
Filippi, J.-B., Mallet, V., & Nader, B. (2014). Representation and evaluation of wildfire propagation simulations. *International Journal of Wildland Fire, 23*(1), 46–57.
Finney, M. A. (1994). Modeling the spread and behavior of prescribed natural fires. In *Proceedings of the 12th conference on fire and forest meteorology* (pp. 138–143). MD: Society of American Foresters Bethesda.
Finney, M. A. (1998). FARSITE users guide and technical documentation. In *USDA forest Service research paper RMRS-RP-4*. USDA.
Finney, M. A. (2006). An overview of FlamMap fire modeling capabilities. In P. L. Andrews, B. W. Butler, & comps (Eds.), *Fuels management-how to measure success: Conference proceedings. 28-30 March 2006; Portland, OR. Proceedings RMRS-P-41* (Vol. 41, pp. 213–220). Fort Collins, CO: US Department of Agriculture, Forest Service, Rocky Mountain Research Station, 2006.
Gollner, M., et al. (2015). Towards data-driven operational wildfire spread modeling: A report of the NSF-funded WIFIRE workshop. In *Digital repository at the university of Maryland, digital repository at the university of Maryland*. https://doi.org/10.13016/M2Z70R
Hardy, C. C. (2005). Wildland fire hazard and risk: Problems, definitions, and context. *Forest Ecology and Management, 211*(1–2), 73–82.
Leblon, B., Bourgeau-Chavez, L., & San-Miguel-Ayanz, J. (2012). Use of remote sensing in wildfire management. In *Sustainable development-authoritative and leading edge content for environmental management* (pp. 55–81).
Lion, R., Ree, M. M., & Ilja, B. (2002). Priorities in information desire about unknown risks. *Risk Analysis: International Journal, 22*(4), 765–776.
Madey, G. R., et al. (2007). Enhanced situational awareness: Application of DDDAS concepts to emergency and disaster management. In *International conference on computational science*. Berlin, Heidelberg: Springer.
Noonan-Wright, Erin, K., et al. (2011). Developing the US wildland fire decision support system. *Journal of Combustion.* , Article 168473. https://doi.org/10.1155/2011/168473
Pastor, E., et al. (2003). Mathematical models and calculation systems for the study of wildland fire behaviour. *Progress in Energy and Combustion Science, 29*(2), 139–153.
Petrasova, A., et al. (2018). Wildfire spread simulation. In *Tangible modeling with open source GIS* (pp. 155–163). Springer International Publishing.
Richards, G. D. (1990). An elliptical growth model of forest fire fronts and its numerical solution. *International Journal for Numerical Methods in Engineering, 30*(6), 1163–1179.
Richards, G. D. (1995). A general mathematical framework for modeling two-dimensional wildland fire spread. *International Journal of Wildland Fire, 5*(2), 63–72.
Rochoux, M. C., et al. (2014). Towards predictive data-driven simulations of wildfire spread–Part I: Reduced-cost Ensemble Kalman Filter based on a Polynomial Chaos surrogate model for parameter estimation. *Natural Hazards and Earth System Sciences, 14*(11), 2951–2973.
Rochoux, M. C., et al. (2015). Towards predictive data-driven simulations of wildfire spread-Part II: Ensemble Kalman Filter for the state estimation of a front-tracking simulator of wildfire spread. *Natural Hazards and Earth System Sciences, 15*(8), 1721–1739.
Rothermel, R. C. (1972). A mathematical model for predicting fire spread in wildland fuels. In *, Vol. 115. USDA forest Service research paper INT-115, intermountain forest & range experiment station, forest Service*. UT: US Department of Agriculture, Oregon.
Scott, J., & Burgan, R. E. (2005). Standard fire behavior fuel models: A comprehensive set for use with rothermel's surface fire spread model. In *USDA general technical report RMRS-GTR-153*. US Department of Agriculture, Forest Service, Rocky Mountain Research Station.
Sullivan, A. L. (2009). Wildland surface fire spread modelling, 1990–2007. 2: Empirical and quasi-empirical models. *International Journal of Wildland Fire, 18*(4), 369–386.
Sun, Z., et al. (2016). Migrating GIS big data computing from Hadoop to spark: An exemplary study using Twitter. In *2016 IEEE 9th international conference on cloud computing (CLOUD)*. IEEE. https://doi.org/10.1109/cloud.2016.0054. Crossref.
Thompson, M. P., et al. (2012). The science and opportunity of wildfire risk assessment. In *Novel approaches and their Applications in risk assessment* (pp. 99–120). InTech.
Wagner, CE Van (1977). Conditions for the start and spread of crown fire. *Canadian Journal of Forest Research, 7*, 23–34.